

ACADEMIC HARDWARE PRODUCTS (MYDAQ, MYRIO)

Register | Login | Help

BOARD

Search the community

SEARCH

Topic Options

All Forum Topics Next Topic >

✓ Solved! Go to solution

myRIO support in LabVIEW 2020 SP1 and 2022 Q3



softball
Member

01-29-2025 10:36 AM Options ▾



Hi

First a disclaimer. **Don't try this unless you have a backup of your system !**
And another disclaimer. **Don't try this unless you are a seasoned myRIO user !**

This description is probably of no use for the student audience of the myRIO. There is no installer so the modifications cannot be undone. So consider this tinkering a proof of concept demonstration.

I have been wondering why there are no support for the mentioned LabVIEW versions. There is little changed in the myRIO toolkits since the first full release with LabVIEW 2014. LabVIEW 2019 was a transition release with the new *Base Image* concept for CompactRIO.

So, I decided to make a brute force attempt and see what happened. Simply copy everything obviously myRIO related from a working myRIO environment to the unsupported environment. This meant copying files from a LabVIEW 2019 SP1 environment to a LabVIEW 2020 SP1 environment. And likewise for 2021 SP1 to 2022 Q3.

LabVIEW 2022 Q3 asked which environment I would like to use, offering a myRIO environment when started. I had no issues opening and running the usual demo project templates for RT and FPGA, and can even install and select the special high throughput personality.

LabVIEW 2019 SP1 crashed immediately when opened, with a nasty cpp file error (as often experienced in the nineties). If was a *myrio.ini* file in the *modes* folder that caused it. Copying the same file over from the 2021 SP1 installation fixed the crash. And 2019 SP1 then runs as nicely as 2022 Q3.

So which files needs to be copied over. I use a dual pane file manager called Total Commander which makes the copying manageable. It involves copying up to about 4000 files. The copy procedure takes about a hour.

I searched for folders including *myRIO* in the name. I copied all such folders and their sub-folders from the 'LabVIEW 2021' folder to the 'LabVIEW 2022' folder, so they ended up in the same named locations.

I searched for additional files including *myRIO* in the name and copied them over. Most were already copied by the folder copying, of course.

Same copy procedure for LabVIEW 2019 SP1 to LabVIEW 2020 SP1.

Copying as described also copies some splash files over, which will cause LabVIEW 2022 Q3 to claim that it has 2021 myRIO support. It can be corrected as this information is simply read from bmp and png files in the *modes* folder.

Remember to install CompactRIO drivers that supports the actual LabVIEW versions.

Regards



0 KUDOS

REPLY

Message 1 of 5(1,833 Views)

Re: myRIO support in LabVIEW 2020 SP1 and 2022 Q3



Bob_Schor
Knight Of NI

03-27-2025 04:37 PM Options ▼

As it turns out, the myRO Software Toolkit 2021 was released on the same day as LabVIEW 2022 Q3. Having installed at least 4 (maybe >5) versions of the Toolkit since the myRIO was released, I always installed Toolkits on already-existing LabVIEW versions of the same name. After a half-dozen attempts that ended in failure, I called NI and managed to get connected to the developers of the Toolkit.

They said to install the Toolkit on a PC without LabVIEW. It sounded crazy to me, but I tried it. It installed LabVIEW 2021 plus some newer components that probably were part of LabVIEW 2022, but I was able to install LabVIEW 2021 after I installed the Toolkit, and now had a working LabVIEW 2021 system that supported the myRIO Toolkit.

They are still selling the Toolkit, though maybe not for much longer. I haven't heard any rumors of later versions of the Toolkit, so it may be that we will be "tied" to LabVIEW 2021 for programming the myRIO. The down side, of course, is that LabVIEW 2021 only has a year or two until it "turns into a frog"

Bob Schor

P.S. -- for some reason, I failed to post this when I originally wrote it over a month ago (I might have fallen asleep with my computer on my lap). The Good News is that they have extended the period they are making and selling myRIOs (through I believe 2027), The Bad News is they have not released information about further myRIO Toolkits, though I've asked my contacts at NI about this.

Anyway, the Good News is that with "modern LabVIEW" (meaning LabVIEW 2019 and later, for all intents and purposes), the following is relatively "safe" (from the standpoint of protecting your PC and its software), provided you have the proper licenses. If you are serious about working with the myRIO and are happy running LabVIEW 2019 or LabVIEW 2021 (these are the only recent versions that I know will work), you can do the following:

1. Be sure you have licenses ready for LabVIEW 2019 and/or 2021 (maybe also 2022, but maybe not). We're going to build a system with only LabVIEW 2019 and 2021, but you can later add other versions.

- 2. Using NIPM, uninstall all National Instruments Software. I usually do this by opening Control Panel, going to Programs and Features, and choosing "NI Software". This brings up NIPM. Be sure "Products only" is not checked. Click the box labelled "Name" (which selects everything except NIPM) and choose "Remove". It may take many minutes (or an hour or so) to uninstall everything. When it does, reboot.
- 3. When it comes back up, do the same thing again. It should find only NIPM. If so, select it and uninstall it. Reboot.
- 4. One more time. You might find a hundred or so things that preceded NIPM, might as well get these, too.
- 5. Now you are ready to reinstall. If you are going to install LabVIEW 2019 with the 2019 Toolkit, go ahead, using either a "kit" (or USB key with everything) or from the Web (i'd advise downloading the Installer rather than downloading the entire off-line kit). Be sure to include LabVIEW Real-Time, DAQmx, VISA, and CompactRIO. I recommend being "selective" with Additional Items.
- 6. Once LabVIEW 2019 is installed, go find the myRIO Software Toolkit 2019 and install that.
- 7. Here's the trick -- do not install LabVIEW 2021. Instead, install the myRIO Software Toolkit 2021. For reasons I don't understand, there are "elements" of LabVIEW 2022 on the myRIO 2021 kit that prevent installation of the toolkit if you install LabVIEW 2021 first (though it might work if you install both 2021 and 2022, but I didn't try that).
- 8. After the Toolkit is installed, go ahead and install LabVIEW 2021. You'll already have installed parts of it from the Toolkit.
- 9. At this point, you can safely install other versions of LabVIEW. However, to the best of my knowledge, they won't support the myRIO Software Toolkit. Note that doesn't necessarily mean that a program written in LabVIEW 2025 can't "talk to" a myRIO running code developed in :LabVIEW 2021 (hmm -- maybe I should try that in my copious free time ...).

Bob Schor

 0 KUDOS

REPLY

Message 2 of 5(1,693 Views)

Re: myRIO support in LabVIEW 2020 SP1 and 2022 Q3 🗨

 softball
Member

Author

03-28-2025 11:17 AM - edited 03-28-2025 11:24 AM Options ▼

Hi

Bob responded with interesting observations.

Yes, myRIO has gotten a little longer life. But apparently only the 1950 version without WIFI :

<https://www.ni.com/en/shop/engineering-education/portable-student-devices/myrio-student-embedded-dev...>

Last Time Buy, April 2026; NI Recommends Transitioning to the NI Single-Board RIO, CompactRIO Hardware, or Digilent FPGA

NI will continue to support myRIO until July 2028. For embedded education and projects, NI recommends Single-Board RIO and CompactRIO Hardware.

For FPGA and digital systems courses, consider Digilent FPGA on digilent.com.

If myRIO specifications are required, the myRIO-1950 Student Embedded Design version will be available until December 2027.

Bob states that the myRIO Toolkit is 'sold'. Well, it is actually free, but not of any use in itself without the RT software (and FPGA software) which is certainly not free.

Bob also told about his frustrations installing the recent myRIO related software. On several occasions.

I have no current issues installing software for myRIO for 2019 and newer. I was confused at one point as I didn't take notice of the new **base image** concept where a minimal base image is supposed to be installed on the myRIO, which currently supports 2019, 2020, 2021 and 2022. With regards to myRIO the concept is unfinished work, as there is no myRIO Toolkit for neither 2020 nor 2022 Q3. So even if you can install CompactRIO firmware support for 2020 and 2022, you can't use it in LabVIEW without the missing myRIO Toolkit.

I described my hackery to get support for myRIO in all four version of LabVIEW (2019 through 2022 Q3). Currently there is no never CompactRIO software than this supporting myRIO. My hackery by the way works splendidly.

I guess NI ran out of time to support LabVIEW 2022 Q3 with a proper myRIO Toolkit when releasing LabVIEW 2022 Q3. Instead they resorted to 'secretly' install LabVIEW 2021 software, if you selected myRIO Toolkit in the 2022 Q3 installer.

It works, but is a dubious practice. Especially as Ni never mentioned that trickery anywhere.

Also repeating myself, I advice against installing multiple versions of LabVIEW on the same partition if you need to install any dirver, toolkit or module. You will regret it when it doesn't work and you need to repair something.

Regards

★

1 KUDO

REPLY

Message 3 of 5(1,682 Views)

Re: myRIO support in LabVIEW 2020 SP1 and 2022 Q3



15k

Bob_Schor



Knight Of NI

03-29-2025 10:13 AM Options ▾

Ouch! I didn't realize they were only selling/supporting the 1950! The 1900 model is the one we use -- having wireless makes a very nice difference, especially when you "build it in" as the "smarts" to another piece of equipment.

Bob Schor

★

0 KUDOS

REPLY

Message 4 of 5(1,665 Views)

 Solution

Accepted by topic author softball

 Re: myRIO support in LabVIEW 2020 SP1 and 2022 Q3



 softball

Member

Author

06-15-2026 10:07 AM - edited 06-15-2026 10:29 AM Options ▾

Hi

Google found this web site, posted April 2026 :

<https://kreiseder.org/2026/04/using-the-myrio-toolkit-beyond-labview-2021-a-practical-patch-for-labv...>

This German site describes the same 'brute' procedure for updating that I have already outlined.

But they added a full description and images. Very nice initiative.

They even added an option to mail them to get what is needed. I have not tried that.

They had success installing the myRIO into LabVIEW 2026 and 2025. Quote :

We **have tested** the workaround with the versions **LabVIEW 2026 Q1 (64 bit)** and **LabVIEW 2025 Q3 (64 bit)**. It is to be assumed that this workaround will also work for LabVIEW 2022, 2023 and 2024. Furthermore **brilliant**, it runs in 32 bits and 64 bit version of LabVIEW.

So, now a myRIO can be fully supported in even the newest version of LabVIEW.

Regards

PS : Prior to this new knowledge, I had also tested my 'brute' method with LabVIEW 2023 Q1 32 bits using the same procedure as for LabVIEW 2022 Q3. It worked fine. Something new was that each CompactRIO version after 2022 Q3 now seems to include and require a new firmware for myRIO to replace the long living version 8.8.



0 KUDOS

REPLY

Message 5 of 5(95 Views)

 All Forum Topics Next Topic >

Powered by
Khoros 

Solutions	Orders	Company	Support
Academic & Research	NI Distribution Partners	NI is now part of Emerson	Downloads
Aerospace, Defense, & Government	Order Status and History	About	Product Documentation
Electronics	Retrieve a Quote	Emerson Careers	Discussion Forums
Energy	Terms of Service	Newsroom	Activate a Product
Industrial Machinery	Order by Part Number or Request a Quote	Supply Chain & Quality	Submit a Service Request
Life Sciences		Events	Site Feedback
Semiconductor			
Transportation			

 SOUTH AFRICA (ENGLISH) [LEGAL](#) | [IMPRINT](#) | [PRIVACY](#) | [MANAGE COOKIES](#) © 2026 NATIONAL INSTRUMENTS CORP. ALL RIGHTS RESERVED.